Probability - 2026 final exam

Q1: $x(t) = 2\pi A \cos(t+\theta)$, if $\theta \sim U(0, \pi)$
find $E(x(t))$ or mean:

A. $-\sin t$
B. $-2\sin t$
C. $-4\sin t$
D. $2\sin t$
E. none

Q2: for $X \sim N(1, 1)$ find $P(X > 1) =$

A. 0.25
B. 0.5
C. 0.4
D. 0.6
E. none

Q3: $\int_{-\infty}^{\infty} \sin x \; \delta(x+\pi)\, dx =$

A. 0
B. -1
C. -0.5
D. 1
E. none

Q4: if $X \sim P(m)$ & $P(X=0) = e^{-4}$, find the value of m:

A. 0
B. 2
C. 3
D. 4
E. none

Q5: $F(0) = 0$, $P(0 < x < a) = a$. Find $F(F(a))$

A. a
B. a^2
C. $2a$
D. a^3
E. none

Q6: For the independent random variables x and y where $EX > 0$ and $EXY = 4EX$, find EY:

A. -2
B. 2
C. 4
D. -4
E. none

Q7: X & Y are orthogonal r.vs,
$R_{xy} = 2 + E(x(t))$, $E(x(t)) =$

A. 2

B. -2

C. 1

D. -1

E. none

Q8: Two random variables X & Y are related $y = a^2x + b$, (a, b) are real numbers then $Cov(X, Y)$ is :

A. $a^2\sigma^2$

B. $a\sigma$

C. aEX^2

D. $a(EX)^2$

E. none

Q9: $X = \{1, 2\}$ with equil probabilites and $Y = X - 1$, find

a. EX

b. EX^2

c. EY

d. EXY

Q10: For $f(x, y) = 1$, find the probability that x & y are less than 0.5.

Q11: If $X \sim B(n, p)$, prove that $\text{Var } X \leq EX$ is always true:

Q12: For a random process with zero mean, the variance is:

Q13: Is $x(t) = e^{-t\phi}$ a wide-sense stationary for $\phi \sim U[0,1]$?

Q14: If $C_{xx}(\tau) = R_{xx}(\tau)$, find the mean of the process.